

OMER KOC

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Education

Northeastern University

Jan. 2023 – May. 2025

M.Sc. in Data Science. GPA: 3.96

Boston, MA

Relevant Coursework: Algorithms, Unsupervised/Supervised Machine Learning, Large Language Models

Istanbul Technical University

Sep. 2016 – Jun. 2021

B.Sc. in Industrial Engineering. High Honor List. GPA: 3.73

Istanbul, Turkey

Experience

MATREU

Sep. 2024 – Dec. 2024

Data Analyst | *Python, Excel, Amazon, Helium 10, Online Marketing Analysis*

Austin, TX

- Automated data workflows and reporting, increasing efficiency and ensuring timely delivery of insights to the team.
- Analyzed over 50 product performance metrics, identifying opportunities to improve sales and optimize product listings, leading to a 10% increase in monthly revenue.

Khoury College of Computer Sciences

Dec. 2023 – Jun. 2024

Graduate Teaching Assistant | *Python, SQL, Apache Spark, MongoDB, Web Scraping*

Boston, MA

- Held weekly office hours to provide support on SQL, Python, Data Modeling, and tools such as MongoDB and Spark.
- Evaluated student code for accuracy and adherence to requirements and assigned grades for projects and exams.

Dynflux

Jan. 2022 – Dec. 2023

Data Science & Analytics Specialist | *Python, SQL, Power BI, Git, Predictive Modeling*

Boston, MA

- Built predictive models (XGBoost, Random Forest) and automated pipelines (SQL, Pandas) to forecast demand and optimize pricing, boosting sales efficiency by 15%.
- Designed KPI dashboards in Power BI, enabling data-driven decisions across business units.

Turkcell

Dec. 2020 – Jan. 2022

Data Analyst | *Oracle SQL, Python, Excel, Power BI, Reporting*

Istanbul, Turkey

- Streamlined data workflows with Python and SQL, boosting forecast accuracy by 25% and leveraging ML for insights.
- Automated data extraction processes using SQL and Python, reducing manual effort by 30% across reporting tasks.

Selected Projects

Urban Air Quality Forecasting | *Python, XGBoost, LSTM, Tableau* 🐙

May. 2025

- Developed XGBoost and LSTM models to forecast urban air pollutants using 5 years of weather and traffic data.
- Performed feature engineering and deployed an interactive Tableau dashboard to visualize pollution trends across cities.

IMDb Movie Review Sentiment Analysis | *Python, Deep Learning, NLP* 🐙

Dec. 2023

- Built a CNN-LSTM hybrid model for NLP tasks with embeddings and dropout, achieving 90.18% accuracy.
- Preprocessed data with HTML removal, stopword elimination, and lemmatization for consistency.

Design & Implement a Relational Database: Bird Strike | *R, MySQL* 🐙

May. 2023

- Developed a relational database in MySQL for bird strike incidents, incorporating advanced data manipulation with R.
- Developed advanced SQL queries for data extraction and transformation in MySQL.

Intraday Electricity Market Price Forecast Using Machine Learning | *Python, Machine Learning* 🐙

Jun. 2021

- Achieved a \$1.5 MAE using a CNN-LSTM hybrid model, leveraging deep learning techniques to outperform Bi-LSTM and GRU models, demonstrating expertise in time-series analysis.
- Used Facebook's Prophet to preprocess over 50 features, enhancing model accuracy with advanced forecasting techniques.

Leadership / Extracurricular

New England Turkish Student Association (NETSA)

Sep. 2023 – Present

Vice President | *Leadership, Team Coordination, Outreach, Public Speaking, Fundraising*

- Increased the organization's treasury by 45% compared to the previous year through strategic grant and sponsorships.
- Facilitated networking events, strengthening the connection between Turkish and American student communities.

Technical Skills

Languages & Tools: Python, R, SQL, C++, MongoDB, Amazon Web Services (AWS), Microsoft Azure, MS Office

Expertise: Data Analysis, Statistics, Algorithms, Machine Learning, Optimization, Relational Databases, LLM's